Offline Rowing IMU Dashboard - User Guide

Purpose: prepare a SEAT and BOAT IMU recording so stroke detection is plausible and coaches or athletes can understand the most important setup steps. Dashboard speed values are estimated movement curves. They are useful for timing and comparison, but they are not calibrated physical speed in m/s.

Filter note: the current system does not use a Kalman filter. The current pipeline removes a baseline, smooths acceleration with a moving average, estimates a speed curve from acceleration, corrects simple drift, and then detects stroke starts from the local movement curve.

1. Record and open the report

1. Turn on both IMUs and record the session. The SD cards do not need to be connected at the same time later. Copy the CSV files into one folder.

2. Run the offline analysis with that folder as the input. The tool creates a self-contained HTML file. This file can be opened without Python and can be sent to another person.

3. First check whether SEAT and BOAT were detected correctly. If the files were swapped, use swap SEAT/BOAT data in the Sensor alignment section.

2. Sensor alignment

Setting	When to use it	Effect
Forward axis	Use this when the IMU is mounted differently and another axis points along the rowing direction.	Selects the axis used as forward movement. The current default is Y lateral as forward.
invert forward sign	Use this if the curves look logically mirrored.	Flips the displayed movement direction.
invert SEAT only	Use this if only the seat sensor was mounted in the opposite direction.	Flips only the SEAT sensor direction.
swap SEAT/BOAT data	Use this if the CSV files were detected or selected in the wrong order.	Swaps the two data sources inside the dashboard.

3. Stroke detection tuning

The dashboard defines one stroke window from one detected drive start to the next detected drive start. For calibration, inspect about 1-2 minutes of rowing and write down the drive start times: first drive start, second drive start, third drive start, and so on.

After writing down the drive start times, calculate the time between neighboring starts. The smallest of these values is the minimum real time between strokes in this calibration window. Set Stroke detection: minimum gap ms slightly smaller than that value. Keep smoothing samples at 10 for the first test.

Step	Action	Meaning
1	Choose a 1-2 minute calibration window for one athlete.	Use a part of the recording with normal rowing rhythm.
2	Write down each visible drive start time.	Use the table below: drive start 1, drive start 2, drive start 3, etc.
3	Calculate the time gaps between starts.	Example: start 2 - start 1, start 3 - start 2, start 4 - start 3.
4	Find the smallest gap.	This is the shortest real stroke interval in the calibration window.
5	Set minimum gap slightly smaller than the smallest gap.	Example: smallest gap is 1.20 s, use about 1100 ms.
6	Keep smoothing samples at 10 first.	Only change smoothing later if false detections remain.

4. Athlete segments and export

If multiple athletes row one after another, keep everything in one long recording. In the coach dashboard, mark a time range in the graph or type it into the zoom fields, then click Save visible time range as athlete. Each athlete receives a separate segment row.

Before exporting HTML, the coach can write a short message for the athlete. The exported athlete report shows the same main graphs as the coach report, but time is local to that athlete. A segment from 60 to 110 seconds is shown as 0 to 50 seconds in the athlete report.

Use this for calibration

Fill this in for each athlete used for tuning. For the first test, smoothing samples should stay at 10. Minimum gap should be slightly smaller than the smallest time between two neighboring drive starts.

Athlete 1 drive start times

Athlete 1	Drive start 1 (s)	Drive start 2 (s)	Drive start 3 (s)	Drive start 4 (s)	Drive start 5 (s)	Drive start 6 (s)
	Drive start 7 (s)	Drive start 8 (s)	Drive start 9 (s)	Drive start 10 (s)	Drive start 11 (s)	Drive start 12 (s)

Calibration window	Smallest time between starts (s)	Recommended minimum gap (ms)	Smoothing samples	Notes
			10	

Athlete 2 drive start times

Athlete 2	Drive start 1 (s)	Drive start 2 (s)	Drive start 3 (s)	Drive start 4 (s)	Drive start 5 (s)	Drive start 6 (s)
	Drive start 7 (s)	Drive start 8 (s)	Drive start 9 (s)	Drive start 10 (s)	Drive start 11 (s)	Drive start 12 (s)

Calibration window	Smallest time between starts (s)	Recommended minimum gap (ms)	Smoothing samples	Notes
			10	